

Omer Abdur Rehman

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LINKS

<https://omerabdurrehman.github.io/portfolio/>
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PROFILE

A motivated and detail-oriented Computer System Engineering graduate with practical experience in Python, C++, data analysis, machine learning, and application development. Demonstrated ability to design, develop, and optimize technical solutions through analytical problem-solving and hands-on implementation. Proficient in MySQL, Microsoft Excel, Flutter development, and emerging AI technologies, with a strong commitment to continuous learning and delivering high-quality results in fast-paced environments.

Experience

June 2025 – August 2025

I completed a two month internship at the Punjab Information Technology Board (PITB) as a Flutter App Development Intern, contributing to cross platform mobile application development using Flutter. Worked on both frontend and backend implementation, with a focus on responsive UI design, application logic integration, and performance optimization within a professional IT environment.

PROJECTS

Advanced Greenhouse Controller System

- Designed an automated greenhouse monitoring system using the AT89C51 microcontroller to track temperature and humidity in real time.
- Interfaced sensors via ADC0804 for analog-to-digital conversion with live readings on a 16×2 LCD and LED status indicators (optimal, cold, dry).
- Demonstrated proficiency in sensor interfacing, circuit design, and microcontroller-based automation through a cost-effective embedded solution.

Intelligent Traffic Light Control System

- Designed and implemented a dual-Arduino smart traffic controller with I2C master-slave communication for 4-way intersection management.
- Integrated LDR-based automatic night mode, potentiometer-adjustable signal timing, and a non-blocking finite state machine for real-time responsiveness.

Water Level Monitoring and Alert System

- Developed an automated water level monitoring system with hysteresis-based pump control logic to prevent overflow and dry-running.
- Implemented visual and audible alerts and a non-blocking timing architecture for reliable real-time operation.

Fire Alert System Utilizing the AT89C51 Microcontroller

- Built a low-cost real-time fire hazard detection system with LCD status display and buzzer-based emergency alerts.
- Demonstrated expertise in safety-critical embedded application development and microcontroller sensor interfacing.

Final Year Project

Smart Stock Predictor: A Next-Generation AI Platform for Predictive Financial Intelligence

- Designed and developed an AI-driven stock market prediction platform combining machine learning, sentiment analysis, and 19 engineered technical indicators (RSI, MACD, Moving Averages, ADX, ATR) to forecast market trends.
- Built a Random Forest regression backend in FastAPI with time-series cross-validation and lag features to eliminate data leakage and improve prediction reliability.
- Integrated FinBERT and VADER dual-model sentiment pipeline on Yahoo Finance news for real-time sentiment scoring as a predictive feature.
- Developed a Flutter mobile frontend with real-time stock dashboards, portfolio insights, and visual financial reporting across 17 screens.
- Deployed on Render with Firebase Authentication and a parallelized ML pipeline via Render Workflows for optimized performance.

EDUCATION

2022 - 2026	BSc Computer System Engineering, Islamia University of Bahawalpur	Bahawalpur
2019 - 2021	FSC Pre-Engineering, Punjab Group of Colleges	Bahawalpur
2017 - 2019	Matric, Stars High School	Bahawalpur

SKILLS

Python	MySQL
C++	Artificial Intelligence
Data analysis	Machine Learning

**EXTRA-CURRICULAR
ACTIVITIES**

- Joint Secretary-I, Innovation and Commercialization Society (2025-2026)
- Joint Secretary, Engineering Student Society (2023-2024)
- General Secretary , Departmental Club of CSE (2025-2026)